

Biopolymers, and the computational methods used to study them

A bibliometric mapping of the field, with a Latin American and Colombian focus

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Abstract

This report maps the scientific literature on biopolymers and measures how much of it uses computational chemistry, chemoinformatics or artificial intelligence. It then narrows to Latin America and Colombia. The corpus is built from OpenAlex and cross-checked against Lens, and the method layer is classified by reading abstracts rather than by matching keywords. Two results shape everything that follows. Keyword bibliometrics overstates the computational share of the field worldwide, because the vocabulary of biopolymers overlaps that of structural biology; and it understates the share in Latin America, because much regional work is invisible to affiliation matching. Reading the abstracts reverses the ranking of contexts that the keyword counts produce.

1 What this report answers

Three questions, in the order the work addressed them.

1. Where and when is research on biopolymers being done, and by whom?
2. How much of that research uses computational chemistry, chemoinformatics or artificial intelligence, and which methods on which polymers?
3. What does the Latin American and Colombian position look like inside that map?

The thematic scope is deliberately wide: every biopolymer recognised as a subject of scientific study, anchored on the lines this group works in jointly, namely cellulose and nanocellulose, polyhydroxyalkanoates, and lignin.

2 Method

2.1 How the corpus was defined

The core corpus is the field that *identifies itself* as biopolymers: works whose title or abstract carries a biopolymer framing term. This was reached empirically. Blocks for each polymer family were probed first, and every family block intersected with a biopolymer framing turned out to be a subset of the generic string, so the generic string is the union rather than a narrowing.

Two acronym tests decided the wording of every query. The bare acronym PHA returns 56,015 records against 8,422 when bound to its expanded form, and bare DFT 471 against 295 inside the core corpus. Bare acronyms are used nowhere.

Every search string is a versioned file that is never edited once run, so a count can always be traced to the exact text that produced it.

2.2 Sources

Scopus and Web of Science are not used: the institution has no subscription. The triangulation therefore runs on OpenAlex as the base corpus and Lens as an independent index, with Semantic Scholar and Europe PMC recovering abstracts that OpenAlex lacks.

This turned out to be more informative than a comparison of two commercial indexes would have been. Table 1 shows Lens holding most of the world corpus but a far smaller share of the regional one.

Table 1: OpenAlex against Lens, same concept, same window. Lens covers the world corpus well and the regional corpora poorly, so the choice of source matters most exactly where this report focuses.

Block	Region	OpenAlex	Lens	Lens as % of OpenAlex
Core field	World	142,254	129,729	91.2
Core field	Latin America	8,147	5,412	66.4
Core field	Colombia	598	342	57.2
Method layer	World	3,912	3,323	84.9
Method layer	Latin America	211	104	49.3
Method layer	Colombia	17	4	23.5
Cellulose	Latin America	14,222	9,813	69.0
Cellulose	Colombia	897	558	62.2
PHA/PHB	World	22,225	20,241	91.1
PHA/PHB	Latin America	1,444	1,037	71.8
PHA/PHB	Colombia	88	64	72.7
Lignin	Latin America	11,464	8,598	75.0
Lignin	Colombia	810	635	78.4

The overlap is partial in both directions. Across the matched corpora, between 76% and 100% of Lens records are also in OpenAlex, but only between 24% and 75% of OpenAlex records are in Lens. Lens still contributes works OpenAlex lacks, so the union of the two is larger than either. A single-source count of Latin American output is therefore a lower bound, and this is a result rather than a caveat.

2.3 Reading the abstracts

A keyword query cannot tell a molecular dynamics study of a cellulose nanocrystal composite from a molecular dynamics study of haemoglobin allostery, yet only the first is biopolymer materials science. Every abstract in the method layer was therefore classified by a language model: whether the work is materials research or biological-function research, which polymer family, which computational method, and whether experiments are reported. Every call is cached on a hash of the prompt, so the classification is reproducible and re-running it is free.

One correction was applied after inspecting the output rather than trusting it. The first pass offered a catch-all computational category, and it was absorbing factorial designs, response-surface methodology and curve fitting to the authors’ own measurements. That inflated the Colombian computational share to 12.5% against a defensible 3%. The ambiguous bucket alone was re-asked with a question naming the distinction, and statistical design of experiments is now excluded

from every computational count. The first-pass classifications are kept on disk so the correction is auditable.

2.4 Validating the screening

A classification produced by a language model is worth nothing without an agreement figure. 125 abstracts were double-coded by a second, independent reader on a different model, blind to the first coder’s labels. Asking the same model to check its own work would have it repeat its own mistakes and report high agreement with itself.

Agreement is reported as Cohen’s kappa rather than percent agreement, because percent agreement flatters any classification with a dominant class.

Table 2: Agreement between the two independent coders on the random sample.

Dimension	Agreement	Cohen’s κ
materials vs biological	94.0%	0.878
computational at all	92.0%	0.782
method family	88.0%	0.816
polymer family	80.0%	0.767

The distinction the whole correction depends on, materials research against biological function, reaches the highest agreement of the four. On the conventional reading, 0.61–0.80 is substantial and above 0.80 almost perfect.

The two coders also disagree in a consistent direction. On the random sample of 100, the first coder called 79 works computational against the second coder’s 73, a relative over-detection of 8.2%. Five of the seven disputed works were labelled machine learning, and reading them shows why: they are reviews and experimental papers that mention artificial intelligence as a closing perspective without performing any computation. **Every screened share in this report is therefore an upper bound**, and the strict world figure of 1.28% is closer to 1.18% once the bias is applied. This runs the same way as every other correction found here.

3 Results

3.1 Size, growth and geography

Table 3: The field by context. Growth rates use complete years only. The method share here is the *keyword* estimate, corrected in Section 3.2.

Context	Works, all years	2015–2025	Share of world	Growth/yr	Open access
World	145,743	90,931	100.00%	11.4%	43.2%
Ibero-America	13,095	9,445	10.39%	13.7%	58.5%
Latin America	8,162	6,035	6.64%	14.7%	55.5%
Colombia	598	465	0.51%	27.0%	66.7%

The field holds 145,743 works and grows at 11.4% a year. Both regional contexts grow faster than the world, Latin America at 14.7% and Colombia at 27.0%, so the region is closing the relative gap rather than falling behind. Colombia’s rate is computed on a small base and says more about the base than about capacity. Regional output is markedly more open: 66.7% of Colombian work is open access against 43.2% worldwide.

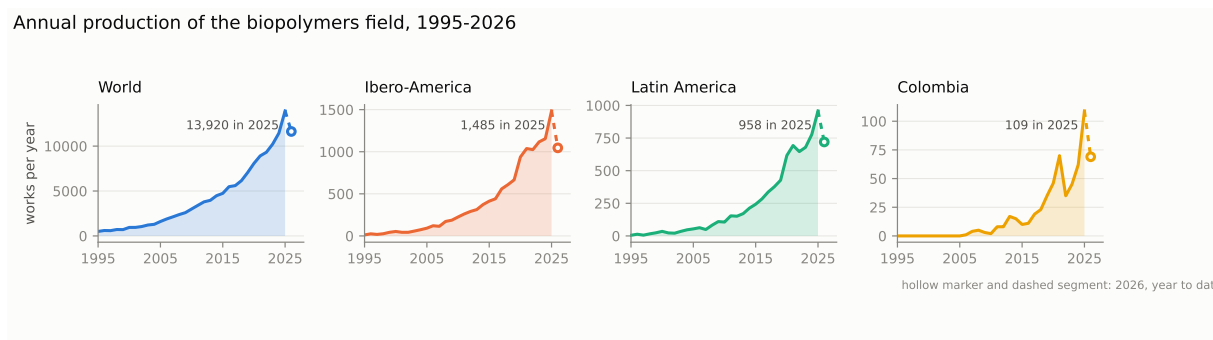


Figure 1: Annual production by context. Each panel carries its own scale, since the contexts differ by three orders of magnitude. The hollow marker and dashed segment are 2026, a partial year.

The top three countries, United States, China and India, hold 50,777 works between them, just over a third of the field. Only one Latin American country reaches the top twenty: Brazil.

Inside Latin America the order by volume is Brazil (4,450), Mexico (1,559), Argentina (794), Colombia (598), Chile (492), Ecuador (232).

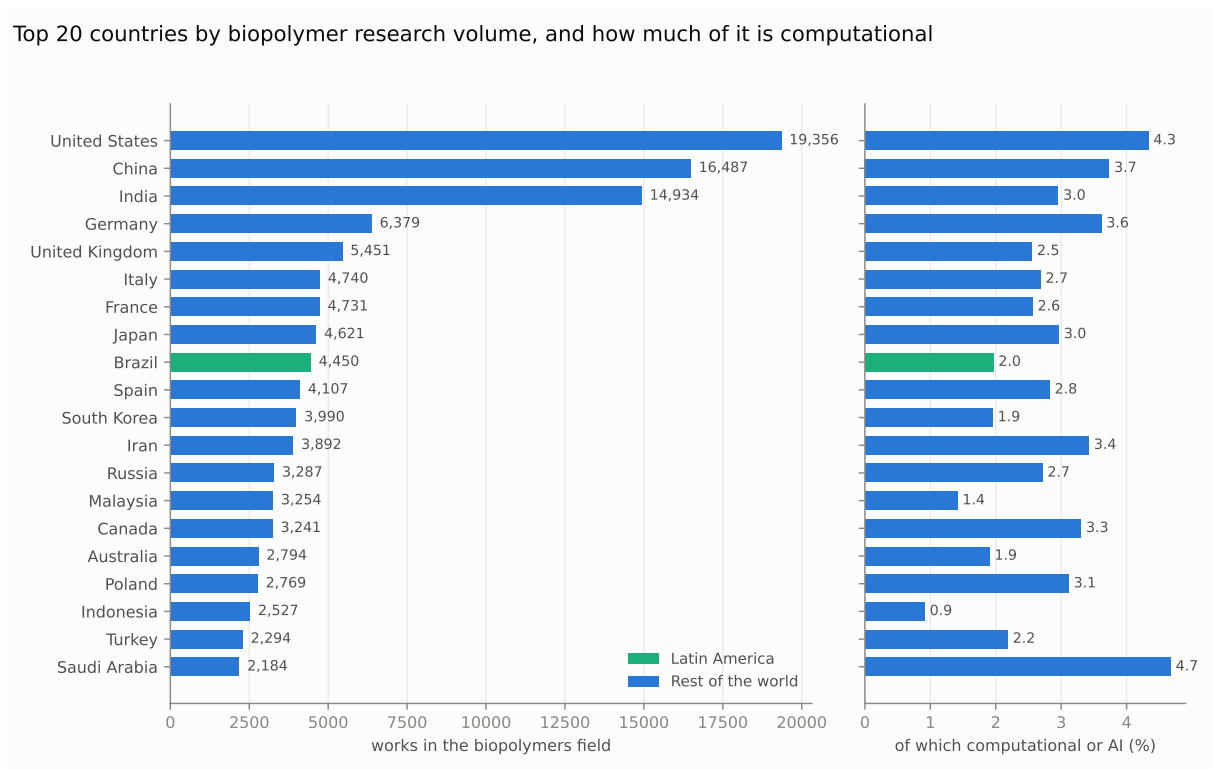


Figure 2: The twenty largest producers, and how much of each country's output is computational. Volume and intensity are separate panels; they are different measures and do not share an axis.

3.2 The computational and artificial-intelligence layer

This is the central result. The keyword estimate puts 2.63% of the field on some computational, chemoinformatic or AI method. Reading the abstracts shows only 56.6% of that layer to be biopolymer materials research; the rest is protein folding, enzyme mechanism and nucleic-acid biology, pulled in because the framing terms overlap. Restricting to materials research brings the share to 1.49%, and restricting further to works with a genuine computational method gives 1.28%.

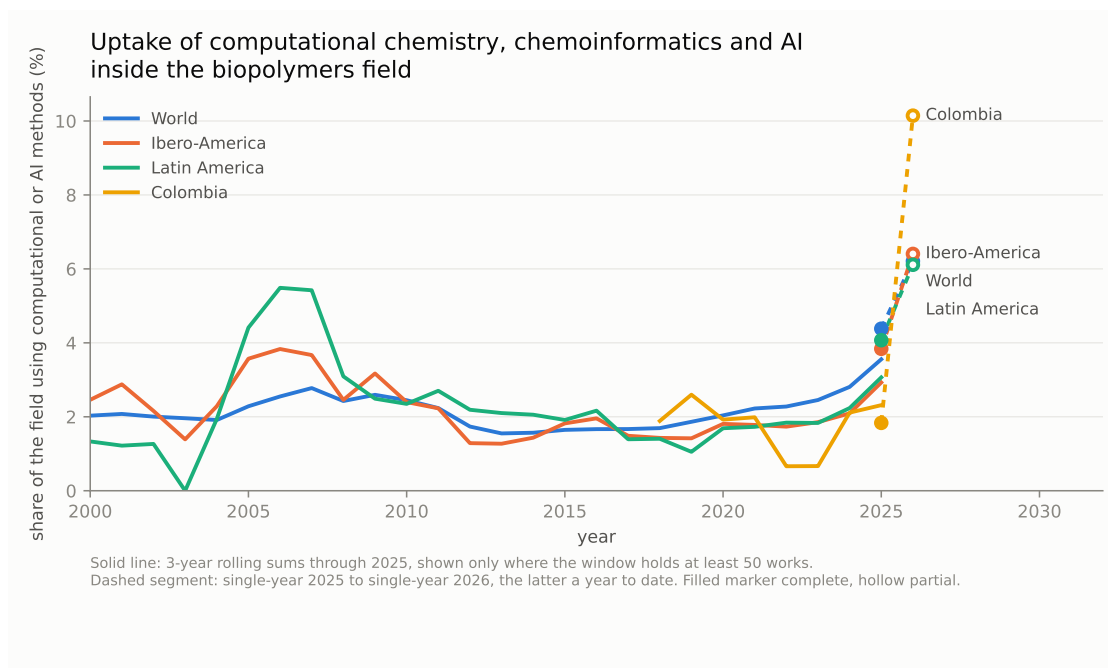


Figure 3: Uptake of computational methods inside the field. The solid line is a three-year rolling share; the dashed segment joins single-year 2025 to single-year 2026, the latter a year to date. Ratios on small denominators are suppressed.

The partial year is not dropped. 2026 is incomplete and is kept, flagged, because it carries the strongest signal in the data.

Table 4: 2026 to date against the whole of 2025. The field as a whole is well short of its 2025 volume at the same date, while the method layer has already passed it in every context.

Context	Whole field			Method layer		
	2025	2026 YTD	%	2025	2026 YTD	%
World	13,920	11,629	84	610	722	118
Ibero-America	1,485	1,046	70	57	67	118
Latin America	958	720	75	39	44	113
Colombia	109	69	63	2	7	350

3.3 Keyword counts understate the region

Table 5: Screening the whole regional field, rather than the keyword-selected method layer, finds computational work the query missed.

Corpus	Materials works	With a method	Screened share	Keyword share
Colombia, whole field	446	20	4.48%	1.94%
Latin America, whole field	5,751	197	3.43%	2.20%

Reading abstracts raises the regional estimate substantially, and it changes the ordering: Colombia sits *above* the Latin American average, not below it as the keyword counts say. The keyword-based ordering should not be quoted.

Part of the reason is mechanical. In the world method corpus, 12.7% of authorships match no institution entity at all, so the country they belong to receives no credit. A 2025 paper with two

Peruvian affiliations and one Mexican one counts, in the data, as Mexican alone. Since every regional corpus here was selected on institution country, works whose regional affiliations all went unmatched never entered it. Per-country loss is worst for the smallest producers. The regional counts in this report are a lower bound.

3.4 Which methods, on which polymers

Table 6: Method mix of the materials-only subset, as a percentage of each context. The world leans on molecular dynamics; the region leans on machine learning and quantum chemistry.

Method	World	Ibero-America	Latin America	Colombia
Molecular dynamics	33.0	28.2	24.2	22.2
Machine learning	24.8	27.5	29.3	22.2
Quantum chemistry	16.5	22.8	24.2	33.3
Docking	5.7	4.7	4.0	0.0
Solubility modelling	3.5	2.0	0.0	0.0
QSAR / QSPR	1.4	0.0	0.0	0.0
Generative	0.6	0.0	0.0	0.0
Other molecular modelling	0.7	2.0	3.0	11.1
Other informatics	0.2	0.7	1.0	0.0
Statistical design of experiments	0.2	0.0	0.0	0.0
None detected	13.4	12.1	14.1	11.1

Generative methods and QSAR are absent from the region entirely, and worldwide the generative slice is a handful of works. That is the emptiest part of the map and the most obvious opening.

3.5 The regional deficit is partly linguistic

Every count so far came from English queries. Spanish and Portuguese term variants were run against the same regional slices and the set difference taken, so what is reported is not how many records the Spanish and Portuguese strings return but how many they return that the English string does *not*.

Table 7: What Spanish and Portuguese terms add. The last column is the set difference: records the Spanish and Portuguese strings return that the English string does not.

Block	Region	English	Spanish/Portuguese	Added
Core field	Latin America	8,159	1,364	1,252 (+15.3%)
Core field	Colombia	598	183	157 (+26.2%)
Cellulose	Latin America	14,669	4,218	3,952 (+26.9%)
Cellulose	Colombia	902	303	281 (+31.1%)
PHA/PHB	Latin America	1,444	221	199 (+13.8%)
PHA/PHB	Colombia	88	36	30 (+34.1%)
Lignin	Latin America	11,527	1,815	1,686 (+14.6%)
Lignin	Colombia	811	158	145 (+17.9%)

The addition is substantial and it is largest for Colombia. A bibliometric count of Latin American biopolymer research built on English terms alone misses between a seventh and a third of the corpus, depending on the slice. Taken with the affiliation-matching loss of Section 3.2, the regional figures in this report understate the region twice over, for two independent and separately measurable reasons.

3.6 The three anchored lines

Table 8: The biopolymer lines this group works in jointly.

Line	World	Latin America	Colombia	Computational, world	Computational, region
Cellulose and nanocellulose	378,840	14,683	902	1.64%	1.69%
PHA / PHB	22,378	1,444	88	1.30%	1.52%
Lignin	181,260	11,534	811	2.08%	1.86%

Lignin is the most computationally studied of the three and PHA the least. Latin America holds a larger share of world output in PHA and lignin than in cellulose, which fits a region working from agro-industrial residues.

The three anchored biopolymer lines

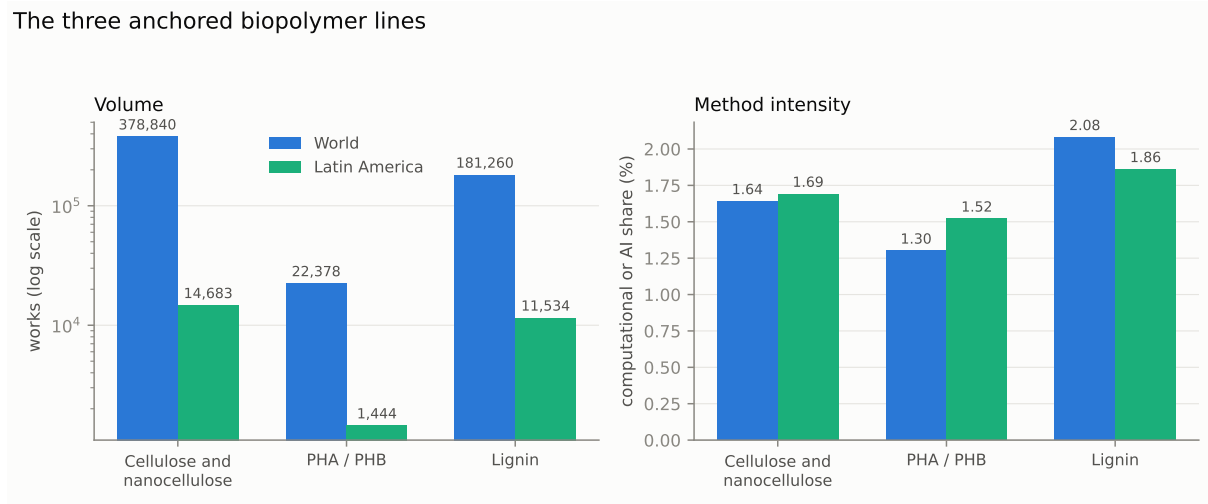


Figure 4: Volume and method intensity of the three anchored lines.

3.7 The generative frontier, read work by work

The screening put ten works in the materials subset on generative or large-language-model methods. Ten is few enough to read every one, and a count cannot say what these works generate, what steers the generation, or whether anything is validated. Both coders read all ten independently. Three disagreements followed, and all three change the answer.

Ten records are nine works. A preprint on retrieval-augmented generation over polyhydroxyalkanoate literature is indexed twice, once with a DOI and once without. The automated pass counted both.

The lignin paper is not a large-language-model paper. It trains autoregressive *molecular* language models on SMILES strings, which is a generative sequence model over chemical structure rather than a language model over natural language. The confusion is understandable and it moves the most methodologically serious work in the set into the wrong category.

Three of the ten sit in venues that are not what they appear to be. Two are published as *Frontiers in Agriculture* and *Frontiers in Chemistry Materials and Catalysis*, under DOI prefix 10.71465, which Crossref registers to *International Study Counselor*. Frontiers Media

SA is 10.3389. A third carries prefix 10.37591, registered to a different publisher again. All three abstracts share a pattern: dramatic headline numbers with no dataset, no baseline and no reproducible protocol. No automated screening asked about venue provenance, and none routinely does.

Table 9: The generative works, as the two readers agreed them. “Specific” marks work genuinely about a biopolymer rather than about polymers in general.

Year	What it generates	Validated by	Specific
2,026	candidate biodegradable polymer formulations	experiment	yes
2,026	candidate biodegradable polymer molecular structures	held out data	yes
2,026	answers about polymer literature	none stated	yes
2,026	candidate polymer repeat units	held out data	no
2,026	polydisperse lignin molecular structures and simulation systems	simulation	yes
2,025	biodegradable polymer structures for food packaging	simulation	yes
2,025	candidate material compositions and microstructures	none stated	no
2,025	3D atomic polymer structures from repeat unit chemistry	held out data	no
2,021	novel HFNAP polymer sequences	experiment	yes

What survives. Removing the works that are not about biopolymers and those whose venue cannot be trusted leaves **four credible biopolymer-specific generative works**, three of them from 2026. Two validated against the physical world, one by electrospinning and mechanical testing, one by measuring nanomolar binding affinities. The electrospun result was honestly negative: the fibre mat was weaker than commercial PET.

That is emptier than a count of ten suggests, and it strengthens the argument of this report rather than weakening it. It also suggests that venue provenance deserves to be checked across the whole corpus, not only on a shortlist, which nothing in the bibliometric literature routinely does.

3.8 Collaboration and community structure

Table 10: Collaboration, counted per work.

Corpus	Works	International	Two or more Latin American countries
World, method layer	3,328	30.4%	0.6%
Latin America, whole field	8,175	34.4%	6.0%
Colombia, whole field	598	44.1%	23.9%
Latin America, method layer	211	45.5%	9.0%

Colombia is the most regionally connected corpus measured. Nearly 24% of Colombian works involve a second Latin American country, against 6% for the region as a whole. Brazil’s strongest ties run outward instead, to the United States, Portugal and Spain, and dwarf its ties inside the region.

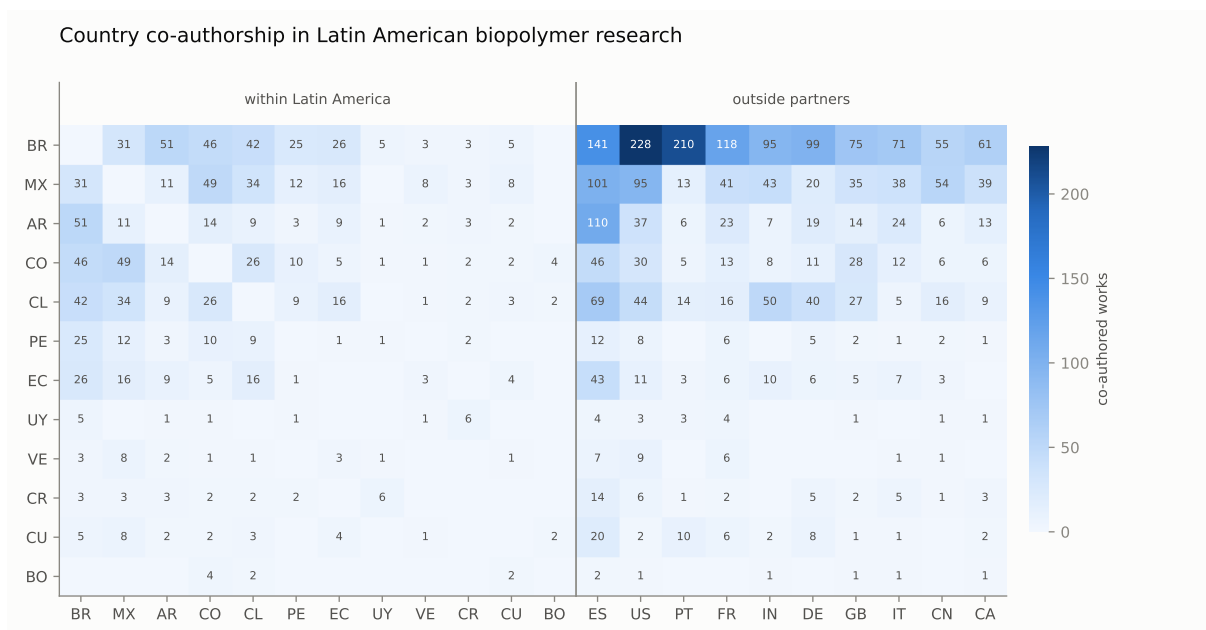


Figure 5: Country co-authorship in Latin American biopolymer research, shown as a matrix. A node-link drawing of the same 115-node graph is a hairball; the network files are provided for VOSviewer instead.

There is no standing computational community. Classic Lotka behaviour gives an exponent near 2. Every corpus here is steeper, and the method layers are steepest of all: of the 14,861 authors in the world method layer, 86.3% appear exactly once, with a fitted exponent of 3.46. The layer is made of one-off contributions. That is what a field looks like before a speciality forms, not after.

And it has no home venue. Bradford’s core zone is the set of journals carrying the first third of a corpus. Worldwide it is 2.4% of journals, which is normal. For the Latin American method layer it takes 13.2% of its journals to reach the first third. That work is scattered, with no venue acting as a centre.

4 Latin America and Colombia

4.1 The claim

Latin America is not behind in computational practice. It is small, invisible and unorganised, and those are three different problems with three different remedies. The region’s difficulty is not access to molecular dynamics or machine learning. It is that its output is thin in absolute terms, that a large part of what exists is not seen by the instruments that count science, and that the part which does use computational methods has neither a community nor a venue.

4.2 Rebuilding the corpus in three languages

Section 3.5 measured what Spanish and Portuguese terms add. The corpus was then rebuilt on the combined vocabulary, and the English-only corpus kept beside it, because the gap between the two is the result.

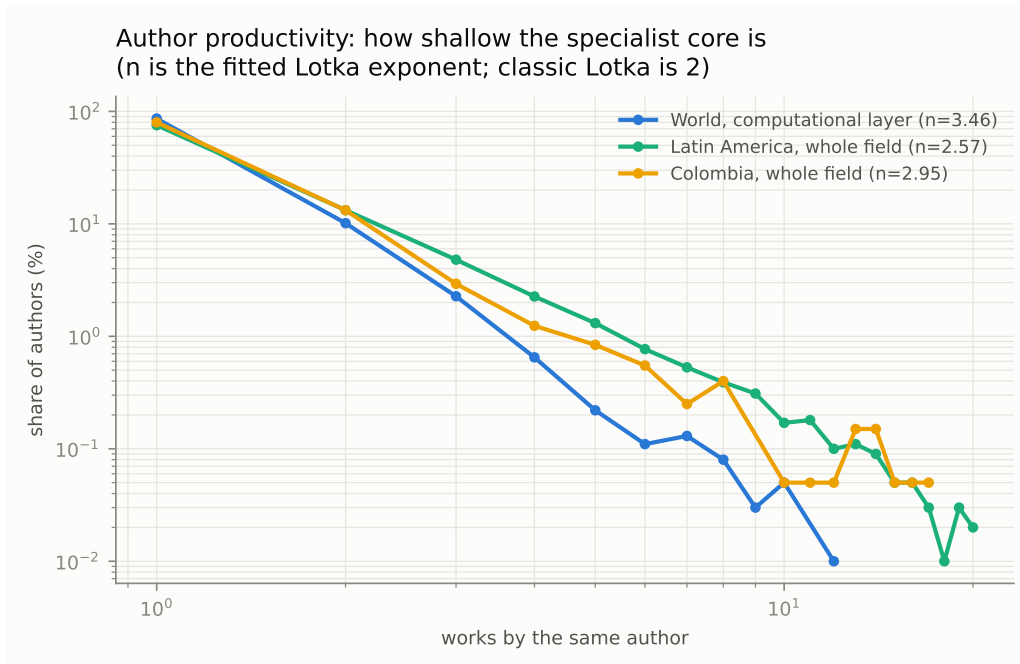


Figure 6: Author productivity. A steeper slope means a shallower core of specialists.

Table 11: What the combined English, Spanish and Portuguese vocabulary adds. The core corpora gain substantially; the method corpora gain almost nothing.

Corpus	English only	Combined	Added
Core field, world	145,743	150,122	4,379 (+3.0%)
Core field, Ibero-America	13,095	14,411	1,316 (+10.1%)
Core field, Latin America	8,162	9,416	1,254 (+15.4%)
Core field, Colombia	598	755	157 (+26.2%)
Method layer, world	3,947	3,960	13 (+0.3%)
Method layer, Ibero-America	327	330	3 (+0.9%)
Method layer, Latin America	211	214	3 (+1.4%)
Method layer, Colombia	17	17	0 (+0.0%)

The result is the opposite of the one expected, and it is the most useful finding of this section. Adding Spanish and Portuguese recovered 7,106 works to the core corpora and 19 to the method corpora. Colombia gains 26.3% of core output and 0.0% of method output.

Confirmed by reading, not only by keyword. A keyword result like this one could be an artefact: English method terms do not fire on Spanish abstracts, so a Spanish-language computational paper would be missed twice over. The screening was therefore re-run on the combined corpora, 14,709 abstracts in all. It confirms the finding and corrects its strength.

Table 12: The screening re-run on the combined corpus. The works recovered by Spanish and Portuguese are markedly less computational than the corpus they join.

Corpus	Materials added	Of which computational	Screened share, English	Combined
Whole field, Colombia	127	2	4.48%	3.84%
Whole field, Latin America	1,114	17	3.43%	3.12%

The recovered literature is 1.5% computational against 4.0% in the corpora it joins: about four tenths as computational, not zero. The keyword match reported 0.0% for Colombia because the English method vocabulary does not fire on Spanish abstracts; reading finds a little, and a little is the right answer.

Two readings follow and they are not exclusive: regional computational work is largely internationalised already, published in English and mostly not invisible; and the Spanish- and Portuguese-language regional literature is predominantly experimental and applied.

The consequence is arithmetic and it cuts against the region. With a complete corpus every measure of regional computational intensity falls, because the denominator grew much faster than the numerator. Colombia’s keyword share goes from 1.94% to 1.59% and its screened share from 4.48% to 3.84%; Latin America’s screened share from 3.43% to 3.12%. The growth rate falls too, Colombia from 27.0% to 18.9%, as older Spanish-language work fills in the early years. **The language correction does not excuse the thin computational layer; it sharpens it.**

Table 13: The field by context, rebuilt on the combined vocabulary. Compare with Table 3, the English-only version.

Context	Works, all years	2015–2025	Share of world	Growth/yr	Method share
World	150,122	93,825	100.00%	11.1%	2.55%
Ibero-America	14,411	10,394	11.08%	12.3%	2.00%
Latin America	9,416	6,944	7.40%	12.4%	1.96%
Colombia	755	566	0.60%	18.9%	1.59%

4.3 Three separate causes of invisibility

A count of Latin American research in this field understates it, and the understatement has three independent sources, each measured rather than assumed. *Language*, worth 15.4% for Latin America and 26.3% for Colombia. *Affiliation matching*, which leaves 12.7% of authorships attached to no institution at all, worst for the smallest producers at 27% for Bolivia and 10% for Venezuela. And *choice of source*, since Lens holds 91% of the OpenAlex world corpus but 57% of the Colombian one. None of these is a statement about the quality of the research. They are statements about the infrastructure that counts it.

4.4 Colombia is the most connected corpus measured

Nearly a quarter of Colombian biopolymer works involve a second Latin American country, four times the regional average, and 44% involve any international partner. Its strongest partners are Mexico, Brazil and Spain in almost equal measure.

Brazil is the contrast. Its strongest ties run outward, to the United States, Portugal and Spain, and dwarf its ties to Argentina or Colombia. The largest producer in the region is the least regionally integrated one. Colombia does not need to be taught to collaborate; it already collaborates more, and more regionally, than the countries above it.

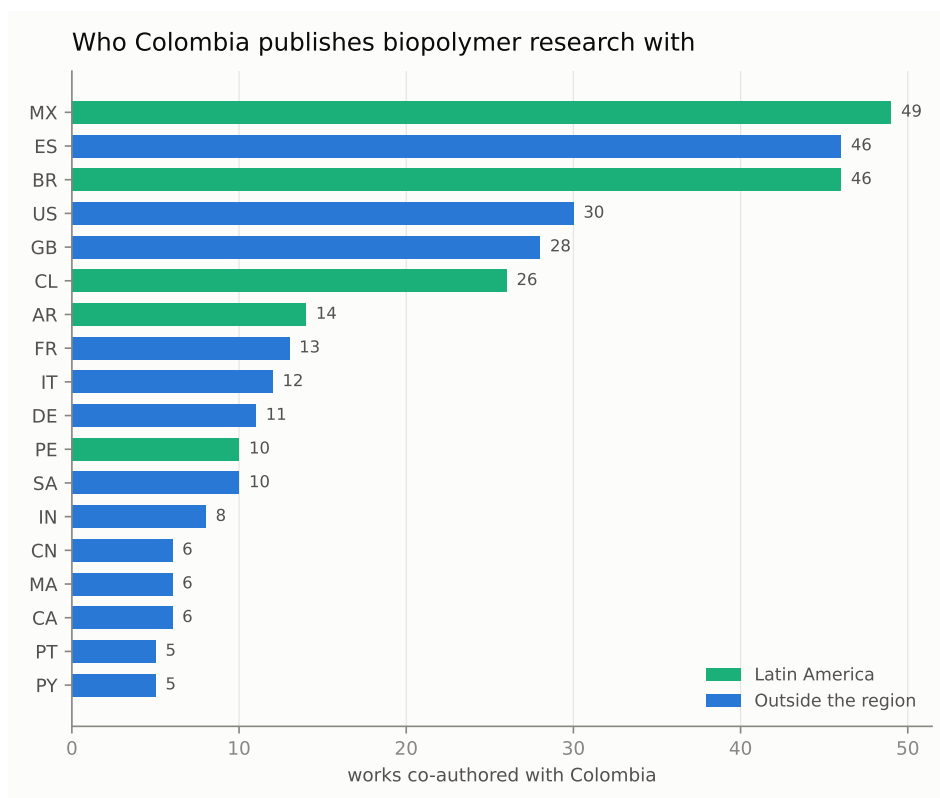


Figure 7: Colombia's co-authorship partners, split between the region and outside it.

4.5 What the region works on

Latin America holds a larger share of world output in lignin and polyhydroxyalkanoates, both 7.1%, than in cellulose at 4.8%. That fits a research base working from agro-industrial residues rather than purified feedstocks, and it is a comparative position rather than a deficiency. On the screened subset Colombia concentrates on chitosan, 22% of its computational work against a world 9%, and on lignin, 11% against 4.7%.

Its institutional base is real but shallow: Universidad Nacional de Colombia, Universidad del Valle, Universidad de Cartagena and Universidad de Antioquia lead, with Universidad de Sucre present at 25 works.

4.6 Three problems, three remedies

Conflating them is how this kind of study usually goes wrong.

1. **Scale is a funding and continuity problem**, not a training one. The region publishes

competently and openly; it publishes little, and the people who do computational work mostly do it once.

2. **Invisibility is an infrastructure problem with identified causes.** Language, affiliation matching and choice of source are each measurable and each fixable, two of them by the databases rather than by the researchers. Depositing with complete, matchable affiliations is the part authors control.
3. **The absence of a community and a venue is an organisational problem,** and it is the one a research group can actually move. A recurring regional venue for computational work on biopolymers would address a gap this study measured rather than assumed.

The clearest opening remains the generative frontier. Reading every generative work in the field leaves four credible biopolymer-specific studies worldwide and none from Latin America (Section 3.7). One of them, on population-aware generation of lignin ensembles, addresses the same problem as this group’s existing lignin line.

5 What the map shows

1. **The computational layer is thin and it is accelerating.** Around one work in eighty in the biopolymers field uses a genuine computational method on a material. That share roughly doubled over a decade and 2026 is running ahead of 2025 in every context measured.
2. **The regional deficit is mostly volume, and the language correction sharpened rather than excused the rest.** On English-only counts Latin America matched the world on method uptake. Rebuilding the corpus in three languages recovered 7,106 works and only 19 of them computational on a reading of their abstracts, so every measure of regional computational intensity falls once the corpus is complete (Section 4.2). Colombia’s screened share goes from 4.48% to 3.84%. The region’s computational work was largely not invisible; its experimental work was. What the region lacks is scale and continuity, and the gap in method uptake is real but modest.
3. **The method mix differs by region.** The world leans on molecular dynamics; Latin America leans on machine learning; Colombia leans on quantum chemistry. An infrastructural reading is available, since sustained dynamics needs computing time that the other two need less of, but the Colombian denominator is small and this belongs in the discussion as a hypothesis rather than in the results as a finding.
4. **Generative methods are essentially absent,** worldwide and entirely so in the region. Reading all of them one by one (Section 3.7) leaves four credible biopolymer-specific works in the world literature. Of everything in this report, that is the clearest opening.
5. **There is no community and no venue.** Most authors in the computational layer publish there once, and the regional slice has no core journals. A thesis that wants to change something has a structural target, not only a topical one.

6 Limitations

- **Two sources, not four.** Scopus and Web of Science were unavailable. OpenAlex and Lens disagree by about a third on regional counts, which bounds how precise any regional figure here can be and is itself reported as a result (Section 4).

- **The screened shares are upper bounds.** The measured over-detection of Section 2.4 is 8.2%, and it is not corrected in the tables, only stated. Read every screened percentage as a ceiling.
- **The world screened shares still rest on the English-only corpus.** The regional corpora were re-screened on the combined vocabulary (Section 4.2); the world method layer gains only 12 works from it, so the world figures move by less than a tenth of a percentage point, but they were not rebuilt.
- **Affiliation matching loses regional work.** Quantified in Section 3.2. The regional counts are a lower bound.
- **Abstract coverage.** Roughly 30% of the method-layer records carry no abstract in OpenAlex. Semantic Scholar and Europe PMC recovered 4,070 of them; the rest could not be read and are excluded from every screened share.
- **The screening is validated, not exhaustively.** 125 abstracts were double-coded by an independent second reader on a different model (Section 2.4), reaching Cohen’s $\kappa = 0.878$ on the dimension the correction depends on, with a measured directional over-detection of 8.2%. That is agreement between two automatic coders, not a hand-coded ground truth; AC has declined a third, human coding pass.
- **English-language queries.** Regional work published in Spanish and Portuguese is undercounted.
- **Country attribution counts a work once per country present,** so country columns sum to more than the corpus.

7 What to do next

1. Extend the combined-vocabulary rebuild to the anchored lines and the world method layer, which still rest on English-only counts.
2. Apply the venue-provenance check of Section 3.7 to the whole corpus. It is a measurable, publishable statement and nothing in the field does it.
3. Take the affiliation-matching loss seriously as a result in its own right. It is a measurable, publishable statement about how invisible small Latin American institutions are to the infrastructure that counts science.